

```
!apt-get update -qq
!apt-get install -y poppler-utils tesseract-ocr
!pip install -q openai openrouter pdfplumber pytesseract pillow pdf2image
```

```
W: Skipping acquire of configured file 'main/source/Sources' as repository 'https://r2u.stat.illinois.edu/ubuntu jammy InRe
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
tesseract-ocr is already the newest version (4.1.1-2.1build1).
The following NEW packages will be installed:
  poppler-utils
0 upgraded, 1 newly installed, 0 to remove and 63 not upgraded.
Need to get 186 kB of archives.
After this operation, 697 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 poppler-utils amd64 22.02.0-2ubuntu0.12 [186 kB]
Fetched 186 kB in 1s (138 kB/s)
Selecting previously unselected package poppler-utils.
(Reading database ... 117540 files and directories currently installed.)
Preparing to unpack .../poppler-utils_22.02.0-2ubuntu0.12_amd64.deb ...
Unpacking poppler-utils (22.02.0-2ubuntu0.12) ...
Setting up poppler-utils (22.02.0-2ubuntu0.12) ...
Processing triggers for man-db (2.10.2-1) ...
```

43.6/43.6 kB	1.6 MB/s	eta 0:00:00
67.9/67.9 kB	4.1 MB/s	eta 0:00:00
290.8/290.8 kB	11.6 MB/s	eta 0:00:00
60.0/60.0 kB	4.1 MB/s	eta 0:00:00
6.6/6.6 MB	65.1 MB/s	eta 0:00:00
3.0/3.0 MB	55.4 MB/s	eta 0:00:00

```
import pdfplumber
import pytesseract
from PIL import Image
from pdf2image import convert_from_path
from google.colab import files
from openai import OpenAI
import os
```

```
client = OpenAI(
    api_key="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", # 🛑 replace
    base_url="https://openrouter.ai/api/v1"
)
```

```
uploaded = files.upload()
file_path = list(uploaded.keys())[0]
print("Uploaded:", file_path)
```

Choose Files No file chosen

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving tata.pdf to tata (1).pdf

```
def extract_text(file_path):
    text = ""

    # ---- Try normal PDF text extraction ----
    if file_path.lower().endswith(".pdf"):
        with pdfplumber.open(file_path) as pdf:
            for page in pdf.pages:
                page_text = page.extract_text()
                if page_text:
                    text += page_text + "\n"

    # ---- If text is too small, use OCR ----
    if len(text.strip()) < 100:
        print("⚠ Scanned PDF detected. Using OCR...")
        images = convert_from_path(file_path)
        for img in images:
            text += pytesseract.image_to_string(img) + "\n"

    return text

# ---- Image files ----
elif file_path.lower().endswith((".png", ".jpg", ".jpeg")):
    img = Image.open(file_path)
    return pytesseract.image_to_string(img)
```

```
# ---- Text file ----
elif file_path.lower().endswith(".txt"):
    with open(file_path, "r") as f:
        return f.read()

else:
    raise ValueError("Unsupported file format")
```

```
report_text = extract_text(file_path)

if not report_text.strip():
    raise ValueError("✗ No text could be extracted from the document")

print("✅ Text extracted successfully")
```

⚠️ Scanned PDF detected. Using OCR...

✅ Text extracted successfully

```
MAX_CHARS = 8000
report_text = report_text[:MAX_CHARS]
```

```
prompt = f"""
You are a medical assistant.
```

```
Tasks:
1. Identify abnormal lab values or findings
2. Explain them in simple language
3. Predict possible diseases or conditions (not diagnosis)
4. Mention severity (mild / moderate / severe)
5. Suggest next medical steps
6. Add a clear disclaimer
```

```
Medical Report:
{report_text}
"""
```

```
response = client.chat.completions.create(
    model="openai/gpt-4o-mini",
    messages=[
        {"role": "system", "content": "You are a careful medical report interpreter."},
        {"role": "user", "content": prompt}
    ],
    temperature=0.2
)

print(response.choices[0].message.content)
```

Abnormal Lab Values or Findings

1. **Hemoglobin**: 11 gms/dl (Normal range: M: 13.0 - 18.0 gms, F: 11.5-16.5 gms/dl)
 - **Explanation**: Hemoglobin is a protein in red blood cells that carries oxygen. A low level can indicate anemia.
 - **Possible Conditions**: Anemia, nutritional deficiencies (like iron or vitamin B12), chronic disease.
 - **Severity**: Mild to moderate.
 - **Next Steps**: Further tests to determine the cause of anemia (e.g., iron studies, vitamin levels).
2. **Absolute Eosinophil Count**: 460 cells/cumm (Normal range: 40-440 cells/cumm)
 - **Explanation**: Eosinophils are a type of white blood cell involved in allergic reactions and fighting parasites. An elevated count may suggest an allergic reaction or infection.
 - **Possible Conditions**: Allergies, asthma, parasitic infections, or certain autoimmune diseases.
 - **Severity**: Mild to moderate.
 - **Next Steps**: Allergy testing or stool tests for parasites.
3. **Serum Calcium**: 8.1 mg/dl (Normal range: 8.4 - 11.5 mg/dl)
 - **Explanation**: Calcium is important for bone health and other bodily functions. Low levels can affect muscle and nerve function.
 - **Possible Conditions**: Hypocalcemia, vitamin D deficiency, or parathyroid issues.
 - **Severity**: Mild.
 - **Next Steps**: Check vitamin D levels and consider dietary adjustments or supplements.

Summary of Findings

- **Hemoglobin** is low, indicating potential anemia.
- **Eosinophil count** is slightly elevated, which may suggest an allergic reaction or infection.
- **Calcium level** is low, which may require further investigation.

Disclaimer

This interpretation is for informational purposes only and should not be considered a diagnosis or medical advice. Please consult a healthcare professional for further evaluation.

